

1. Use Bayes' rule to algebraically prove that, for any events  $A$  and  $B$ ,  $P(A) \geq P(A|B)P(B)$ . Explain in plain language why this result is not surprising.
2. George the Coin Tamperer gives me a randomly-selected coin.  $1/5$  of George's coins will always come up heads; the rest are fair. I flip the coin twice. What is the probability that I get two heads?
3. George the Coin Tamperer is back. He gives me another randomly-selected coin. I flip the coin five times. It comes up heads each times. What is the probability that George gave me a heads-only coin?
4. This year, Renowned University's acceptance rate was 10%. A new test is designed to tell applicants if they'll get into Renowned. 90% of Renowned students can pass the test, but only 18% of all applicants pass it. Assuming that whether a *specific* student or applicant passes is determined randomly, is passing the test a good predictor of whether an applicant will get into Renowned? Is failing the test a good predictor?